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Genetic Variability, Heritability and Trait Association Studies in Forage Oat (*Avena sativa* L.) for Yield and Quality Traits

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Abstract
The present study assessed genetic variability, heritability, and character association among 13 oat (*Avena sativa* L.) genotypes for forage yield and quality traits at G. B. Pant University of Agriculture and Technology, Pantnagar during Rabi 2022-23. Genotypes were evaluated in a Randomized Block Design with three replications for eleven traits. Analysis of variance revealed significant differences among genotypes for all traits. Phenotypic coefficient of variation exceeded genotypic coefficient of variation for all traits, with narrow differences indicating limited environmental influence. High heritability coupled with high genetic advance as percentage of mean was recorded for ADF (97.94%, 30.76%) and crude protein (93.13%, 22.61%), suggesting effectiveness of direct selection. UPO-22-2 was superior for green forage yield (149.17 kg/plot), dry matter yield (33.22 kg/plot), and crude protein (15.16%), while UPO-212 recorded the lowest ADF (7.15%) and NDF (28.60%), indicating higher digestibility potential. Dry matter yield showed the strongest positive correlation (r = 0.624**) and highest direct effect (0.568) on green forage yield, confirming its importance as a primary selection criterion.

Keywords: *Avena sativa*, forage quality, genetic variability, path analysis, correlation, oat genotypes

Introduction
Oat (*Avena sativa* L.) is a valuable winter forage cereal as it has a high level of palatability, digestibility and nutritive value. It is an important component of the dairy industry and has a crucial part to play in satisfying the green fodder needs in the lean winter months in India. In spite of the potential, the productivity of forage oat is still not as high as expected due to the lack of genetic improvement and breadth of variability in the cultivated gene pool [1]. The foundation of any crop-improvement programme is genetic variability which dictates the possibility of selection and genetic gain. This separation of total variability into a genotypic and phenotypic part allows determining traits that are rather determined by genetic influence [2, 3]. High heritability and high advancement in genetics is usually an indicator of additive gene action, which implies that it would be effective to increase it with specific direct selection [4]. Yield in forage crops like oat is a compound quantitative characteristic which depends on a number of morphological and quality parameters. Correlation studies are used to reveal the relationship between these qualities on the other hand, path-coefficient analysis suggested by Wright (1921) [5] and later elaborated by Dewey and Lu (1959) [6] explains the direct and indirect effect that leads to yield formation. The current study was conducted to determine the level of genetic variability, heritability, genetic progression, correlation, and path coefficients of major morphological and quality characteristics that influence the green forage yield in oat genotypes under Pantnagar environment, hence giving a foundation towards the successful selection and breeding of quality and high yielding varieties of oat forage.

Materials and Methods
The experiment was conducted at the Dairy Farm, Nagla, and Seed Testing Laboratory, Department of Genetics and Plant Breeding, G. B. Pant University of Agriculture and Technology, Pantnagar (29.50 N, 79.30 E; altitude 243.84 m) during *Rabi 2022–23*. Experiment was laid out in a Randomized Block Design (RBD) with three replications. Thirteen oat genotypes, including ten test entries and three checks, were evaluated.

Observation was taken for days to 50% flowering(days), plant height(cm), number of tillers per plant, leaf: stem ratio (L/s), days to maturity (days), dry matter yield (kg/plot), dry matter content (%), green forage yield (kg/plot). & Analysis was done for- Crude protein (%), ADF and NDF were estimated as per Goering and Van Soest (1970) [7].

Analysis of Variance (ANOVA)
The data recorded for all morphological and quality traits were subjected to analysis of variance (ANOVA) as per the method suggested by Panse and Sukhatme (1985) [8] under the Randomized Block Design.

Genotypic and Phenotypic Coefficients of Variation (GCV and PCV) [2]

$$GCV(\%) = \frac{\sqrt{\sigma_g^2}}{\bar{X}} \times 100, PCV(\%) = \frac{\sqrt{\sigma_p^2}}{\bar{X}} \times 100$$

Heritability (Broad Sense) [4]

$$h_{bs}^2 = \frac{\sigma_g^2}{\sigma_p^2} \times 100$$

Genetic Advance (GA) and Genetic Advance as Percent of Mean (GAM) [9]

$$GA = k \times \sigma_p \times h_{bs}^2, GAM = \frac{GA}{\bar{X}} \times 100$$

where $k = 2.06$(for 5% selection intensity).

Correlation analysis
The correlation coefficients were estimated using the following expressions [3]:

$$r_{g(xy)} = \frac{g\,Cov(x,y)}{\sqrt{g\,V_x \times g\,V_y}}$$

And

$$r_{p(xy)} = \frac{p\,Cov(x,y)}{\sqrt{p\,V_x \times p\,V_y}}$$

Significance of correlation coefficients was tested using *t*-values at 5% and 1% levels of probability.

Path coefficient analysis
Path coefficient analysis was performed using the method of Wright (1921) [5] and Dewey and Lu (1959) [6], taking green forage yield (GFY) as the dependent variable, and plant height, number of tillers, L/S ratio, DMY, DMC, CP, ADF, and NDF as independent variables.

Results and Discussion
The analysis of variance revealed significant differences among all 13 oat (*Avena sativa* L.) genotypes for all the traits studied, indicating the existence of adequate genetic variability among the evaluated material Table 1. Replication effects were non-significant for all traits, confirming experimental precision. The mean performance of genotypes for all studied traits is presented in Table 2.

Days to 50% Flowering
Days to 50% flowering ranged from 89.01 days (UPO-22-10) to 100.01 days (UPO-22-7), with a grand mean of 93.88 days. UPO-22-10 and UPO-22-4 were the earliest to flower (89.01 and 89.34 days, respectively), which is agronomically desirable as early flowering is generally associated with early maturity and allows timely harvesting of forage. Such variation in flowering time among oat genotypes has been well documented in earlier studies [10].

Plant Height
Plant height showed a range of 137.97 cm (UPO-06-1) to 156.11 cm (UPO-22-6), with a mean of 146.66 cm. UPO-22-6 was the tallest genotype, while UPO-06-1 was the shortest. Taller plants generally contribute to higher biomass accumulation and thus greater forage yield potential. These findings are consistent with earlier reports of significant genotypic variation in plant height in oats [11].

Number of Tillers
The number of tillers per plant varied from 9.68 (UPO-22-2) to 12.01 (UPO-22-6), with a mean of 10.63. UPO-22-6 exhibited the maximum tillering ability. Higher tiller number is a desirable trait in forage crops as it contributes to greater canopy cover and biomass. Similar variation in tiller number among oat genotypes has been reported by earlier workers [12].

Leaf to Stem Ratio
Leaf to stem ratio (LSR) ranged from 0.45 (UPO-212) to 0.65 (UPO-06-1), with a grand mean of 0.55. A higher leaf-to-stem ratio is associated with better forage quality, as leaves are generally richer in protein and more digestible than stems [13]. UPO-06-1 showed the highest LSR, indicating relatively leafier growth.

Days to Maturity
Days to maturity ranged from 124.68 days (UPO-22-10) to 134.68 days (UPO-22-1). UPO-22-10 was the earliest maturing genotype, which can be advantageous in regions where early harvest is desirable. The range of maturity observed is in agreement with earlier studies on oat genotypes [10].

Green Forage Yield
Green forage yield per plot ranged from 102.84 kg/plot (UPO-10-2) to 149.17 kg/plot (UPO-22-2), with a mean of 132.66 kg/plot. UPO-22-2 produced the highest green forage yield, followed by UPO-22-9 (145.23 kg/plot) and UPO-22-5 (141.95 kg/plot). These differences among genotypes for green forage yield have been reported in previous studies [14].

Dry Matter Yield
Dry matter yield ranged from 23.55 kg/plot (UPO-10-2) to 33.22 kg/plot (UPO-22-2), with a mean of 28.04 kg/plot. UPO-22-2 was consistently superior in both green forage and dry matter yield. Dry matter yield is a critical parameter for assessing the productivity of forage crops and its variation among oat genotypes has been reported in previous studies [15].

Dry Matter Content
Dry matter content (%) ranged from 18.11% (UPO-22-9) to 21.63% (UPO-22-6), with a grand mean of 20.16%. The variation in dry matter content reflects the differences in the inherent water content and structural composition of the genotypes [16].

Crude Protein

Crude protein content ranged from 11.33% (UPO-22-1) to 15.16% (UPO-22-2), with a mean of 13.05%. UPO-22-2 registered the highest crude protein content, making it nutritionally superior. High crude protein content is a key quality parameter in forage crops and similar variation has been reported in oat genotypes evaluated for forage quality [17].

Acid Detergent Fibre (ADF) and Neutral Detergent Fibre (NDF): ADF content ranged from 7.15% (UPO-212) to 11.67% (UPO-22-5), with a mean of 9.71%. NDF ranged from 28.60% (UPO-212) to 33.28% (UPO-22-10), with a mean of 30.51%. Lower ADF and NDF values are generally associated with higher digestibility and better forage quality [18]. UPO-212 exhibited low values for both ADF and NDF, suggesting relatively higher digestibility potential.

Table 1: Anova Summary

Sl. No.	Source	Mean Sum of Squares		
		Replication	Treatment	Error
	Degrees of freedom	2	12	24
1	Days to 50% flowering (days)	22.2530	35.816**	11.149
2	Plant height (cm)	53.0860	99.344**	26.532
3	Number of tillers	0.9340	1.437**	0.464
4	Leaf to stem ratio (L/S)	0.0002	0.009**	0.001
5	Days to maturity (days)	16.1120	26.98**	8.094
6	Green forage yield (kg/plot)	2.0130	407.306**	94.054
7	Dry matter yield per plant (kg/plot)	3.230	20.028**	2.591
8	Dry matter content (%)	0.1340	4.054**	1.234
9	Crude Protein (%)	0.0020	6.767**	0.163
10	Acid Detergent Fibre (ADF)%	0.0150	6.479**	0.045
11	Neutral Detergent Fibre (NDF)%	1.0020	7.147**	0.801

* Significant at 5 percent level of significance ** Significant at 1 percent level of significance

Table 2: Mean performance for all 13 genotypes for the assessed characters

Genotypes	Days to 50% flowering (days)	Plant height (cm.)	Number of tillers	Leaf to stem ratio (L/S)	Days to maturity (days)	Green forage yield (kg/plot)	Dry matter yield (kg/plot)	Dry matter content (%)	Crude Protein (%)	Acid Detergent Fibre (ADF)%	Neutral Detergent Fibre (NDF%)
UPO-22-1	94.007	145.013	10.677	0.56	134.677	137.89	30.26	19.003	11.333	9.89	32.547
UPO-22-2	96.68	151.64	9.677	0.59	133.343	149.17	33.217	21.037	15.163	11.453	29.457
UPO-22-3	93.677	153.963	10.343	0.447	129.677	133.08	27.007	19.827	11.55	9.167	29.14
UPO-22-4	89.343	147.223	10.013	0.513	128.34	127.967	28.26	19.91	14.607	8.797	28.993
UPO-22-5	92.01	142.837	11.01	0.573	126.677	141.95	27.71	18.69	12.233	11.667	31.76
UPO-22-6	95.007	156.107	12.013	0.533	131.68	132.96	29.34	21.633	14.923	11.153	31.937
UPO-22-7	100.01	152.167	10.343	0.573	129.673	140.42	31.58	21.58	14.32	9.997	29.75
UPO-22-8	95.677	138.76	10.007	0.543	126.007	125.82	26.967	20.493	12.587	8.237	28.973
UPO-22-9	94.343	148.797	10.343	0.517	128.343	145.23	27.553	18.11	11.717	7.39	31.297
UPO-22-10	89.01	147.22	11.01	0.54	124.677	128.93	27.597	21.113	11.613	10.337	33.28
UPO-212	90.34	143.65	11.677	0.46	127.68	125.417	26.12	20.917	14.587	7.153	28.603
UPO-10-2	91.343	141.173	10.013	0.61	129.01	102.84	23.55	20.81	11.457	10.673	31.003
UPO-06-1	99.01	137.967	11.01	0.65	125.347	132.93	25.3	18.99	13.537	10.267	29.843
Mean	93.88	146.66	10.63	0.55	128.86	132.66	28.04	20.16	13.05	9.71	30.51
CV	6	5.5	6.41	6.84	6.75	7.31	5.74	5.51	3.09	2.19	2.93
SEm	1.93	2.97	0.39	0.02	1.64	5.6	0.93	0.64	0.23	0.12	0.52
CD at 5%	5.63	8.68	1.15	0.06	4.79	16.34	2.71	1.87	0.68	0.36	1.51
CD at 1%	7.63	11.76	1.56	0.09	6.5	22.15	3.68	2.54	0.92	0.49	2.04
Minimum	89.01	137.97	9.68	0.45	124.68	102.84	23.55	18.11	11.33	7.15	28.6
Maximum	100.01	156.11	12.01	0.65	134.68	149.17	33.22	21.63	15.16	11.67	33.28
Replication	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS
Treatment	S	S	S	S	S	S	S	S	S	S	S

Genetic Variability, Heritability, and Genetic Advance

The estimates of genotypic coefficient of variation (GCV), phenotypic coefficient of variation (PCV), broad-sense heritability (h^2), genetic advance (GA), and genetic advance as percentage of mean (GAM) are presented in Table 3. Both GCV and PCV were estimated for all traits, and in all cases PCV was higher than GCV, indicating the influence of environmental effects on phenotypic expression.

The GCV ranged from 1.95% (days to maturity) to 15.09% (ADF%), while PCV ranged from 2.94% (days to maturity) to 15.25% (NDF%). The narrow difference between GCV and PCV for most traits indicates that phenotypic values are a reasonably good reflection of the genotypic values, suggesting limited environmental influence. Similar observations have been made in earlier variability studies on oats [12].

Heritability in broad sense was highest for ADF% (97.94%), followed by crude protein (93.13%), NDF% (72.55%), dry matter yield (69.16%), and leaf to stem ratio (65.78%). Moderate heritability was observed for days to maturity (43.75%), dry matter content (43.25%), number of tillers (41.15%), and days to 50% flowering (42.45%). High heritability estimates indicate that selection based on phenotypic performance would be effective for these traits, as the greater part of the observed variation is of genetic origin

[19]. Similar high heritability for quality traits in oats has been reported by previous workers [11].

Genetic advance as percentage of mean (GAM) was highest for ADF% (30.76%), followed by NDF% (22.61%). Leaf to stem ratio (15.85%) and green forage yield (11.51%) also recorded relatively higher GAM values. The combination of high heritability and high GAM for ADF% and NDF% indicates that selection based on phenotypic performance would be effective for improving these traits [3]. Traits exhibiting high heritability coupled with low GAM, such as days to maturity and plant height, suggest that the realizable genetic gain through selection may be limited for these characters [20].

Table 3: Estimate of variability, heritability and genetic advance

Sl. No.	Genetic Parameters	GCV	PCV	h2(Broad Sense)	GA	GAM
1	Days to 50% flowering (days)	3.05	4.69	42.447	3.85	4.10
2	Plant height (cm)	3.36	4.86	47.774	7.02	4.78
3	Number of tillers	5.36	8.36	41.153	0.75	7.08
4	Leaf to stem ratio (L/S)	9.49	11.70	65.783	0.09	15.85
5	Days to maturity (days)	1.95	2.94	43.75	3.42	2.65
6	Green forage yield (kg/plot)	7.70	10.62	52.611	15.27	11.51
7	Dry matter yield (kg/plot)	8.60	10.34	69.163	4.13	14.73
8	Dry matter content (%)	4.81	7.31	43.248	1.31	6.51
9	Crude Protein (%)	11.37	11.78	93.125	2.95	22.61
10	Acid Detergent Fibre (ADF)%	15.09	15.25	97.938	2.99	30.76
11	Neutral Detergent Fibre (NDF%)	4.77	5.60	72.547	2.55	8.37

GCV – *Genotypic Coefficient of Variation*, PCV – Phenotypic Coefficient of Variation,
h^2 (Broad Sense) – Heritability in Broad Sense, GA – Genetic Advance,
GAM – Genetic Advance as Percent of Mean

Correlation Analysis

Genotypic and phenotypic correlation coefficients between green forage yield (GFY) and other traits, along with inter-trait correlations, are presented in Table 4. In general, genotypic correlations were higher in magnitude than their corresponding phenotypic counterparts, suggesting that the genetic association between traits was somewhat masked by environmental influences a finding consistent with earlier reports in oats and related forage crops [21].

Dry matter yield (DMY) showed a significant positive genotypic (r = 0.624**) and phenotypic (r = 0.634**) correlation with green forage yield, indicating that genotypes with higher dry matter accumulation also tend to produce higher green forage. This strong association suggests that DMY can serve as a reliable indirect selection criterion for improving GFY [21].

Number of tillers (NOT) exhibited a significant positive phenotypic correlation with green forage yield (r = 0.333*), while the genotypic correlation (r = 0.326*) was also positive and significant, suggesting that genotypes with more tillers tend to produce higher forage biomass. This relationship is biologically logical as tillers directly contribute to crop canopy and biomass accumulation [22].

Plant height showed a positive but non-significant genotypic (r = 0.282) and phenotypic (r = 0.296) correlation with GFY. Leaf to stem ratio (r = 0.241 genotypic; r = 0.242 phenotypic) and crude protein (r = 0.188 genotypic; r = 0.188 phenotypic) also showed positive but non-significant associations with GFY.

Dry matter content (DMC) showed a negative association with GFY at both genotypic (r = −0.234) and phenotypic (r = −0.235) levels. ADF and NDF showed negligible correlations with GFY, which is expected since fibre content is more closely linked to quality parameters than to yield per se.

Among inter-trait correlations, DMY showed a significant positive genotypic correlation with plant height (r = 0.506*) and crude protein (r = 0.368*). Crude protein was significantly and positively correlated with DMC at genotypic level (r = 0.429*) but negatively with NDF% (r = −0.436* genotypic; r = −0.439* phenotypic), indicating that genotypes with higher crude protein tend to have lower fibre fractions a well-established inverse relationship in forage quality [23]. Leaf to stem ratio showed a significant positive correlation with ADF% at both genotypic (r = 0.514**) and phenotypic (r = 0.521**) levels.

Table 4: Combined Genotypic (Above Diagonal) and Phenotypic (Below Diagonal) correlation table for traits

Genotypic and phenotypic Correlation Matrix									
	PH (cm)	NOT	LSR (L/S)	DMY (kg/plot)	DMC (%)	CP (%)	ADF%	NDF%	GFY (kg/plot)
PH (cm)	1	0.138	-0.233	0.506*	0.176	0.185	0.102	0.099	0.282
NOT	0.108	1	0.061	0.242	-0.062	0.223	0.079	0.247	0.326*
LSR (L/S)	-0.220	0.084	1	0.294	-0.026	0.123	0.514**	0.259	0.241
DMY (kg/plot)	0.498*	0.233	0.301	1	0.115	0.368*	0.255	0.105	0.624**
DMC (%)	0.190	-0.059	-0.023	0.125	1	0.429*	0.143	-0.086	-0.234
CP (%)	0.190	0.225	0.126	0.373*	0.429*	1	0.069	-0.436*	0.188
ADF%	0.109	0.082	0.521**	0.260	0.142	0.069	1	0.379*	0.08
NDF%	0.084	0.246	0.256	0.088	-0.079	-0.439*	0.384*	1	0.052
GFY (kg/plot)	0.296	0.333*	0.242	0.634**	-0.235	0.188	0.079	0.054	1

Note – PH = plant height, NOT = No. of tillers, LSR = leaf stem ratio, DMY = dry matter yield, DMC = dry matter content, CP = crude protein, ADF = Acid detergent fibre, NDF = Neutral detergent fibre, GFY = Green forage yield. (* Significant at five percent level of significance, ** Significant at one percent level of significance)

Path Coefficient Analysis

Path coefficient analysis was carried out to partition the genotypic and phenotypic correlation coefficients of various traits with green forage yield into direct and indirect effects, providing a clearer understanding of the cause-and-effect relationships (Tables 5 and 6).

Dry matter yield (DMY) exerted the highest positive direct effect on green forage yield at both genotypic (0.5681) and phenotypic (0.5695) levels. This, combined with a high positive correlation with GFY, confirms that DMY is the most important trait directly influencing forage yield and should be given primary consideration in selection programmes. Similar findings have been reported in oats and other forage crops [21].

Number of tillers (NOT) had a moderate direct positive effect on GFY at genotypic (0.1797) and phenotypic (0.1840) levels, with additional indirect contributions through DMY. Plant height also showed a positive direct effect (genotypic: 0.0693; phenotypic: 0.0954), with a large portion of its correlation with GFY being mediated indirectly through DMY (indirect effect: 0.2875 genotypic).

Leaf to stem ratio (LSR) had a moderate positive direct effect (genotypic: 0.1352; phenotypic: 0.1360) on GFY. Its indirect effect through DMY (0.1670 genotypic) and ADF% (−0.0415 genotypic) contributed to its overall genotypic correlation with GFY.

Dry matter content (DMC) exhibited a negative direct effect on GFY at both genotypic (−0.2942) and phenotypic (−0.3047) levels. Similarly, ADF% (genotypic: −0.0807; phenotypic: −0.0962) and NDF% (genotypic: −0.0873; phenotypic: −0.0687) exhibited negative direct effects. The negative direct effects of fibre fractions on GFY are consistent with their role as quality-reducing components in forage, and have been noted in previous studies [15].

Crude protein (CP) had a very small positive direct effect on GFY (genotypic: 0.0031; phenotypic: 0.0061), with its correlation being primarily mediated through DMY (indirect effect: 0.2091 genotypic). This indicates that the positive association between CP and GFY is largely indirect rather than causal.

Overall, the path analysis underscores the importance of dry matter yield and number of tillers as the primary selection criteria for enhancing green forage yield in oat breeding programmes.

Table 5: Genotypic Path Coefficients for traits

	PH (cm)	NOT	LSR (L/S)	DMY (kg/plot)	DMC (%)	CP (%)	ADF%	NDF%	GFY (kg/plot)
PH (cm)	+0.0693	+0.0248	-0.0315	+0.2875	-0.0518	+0.0006	-0.0082	-0.0086	+0.2820
NOT	+0.0096	+0.1797	+0.0082	+0.1375	+0.0182	+0.0007	-0.0064	-0.0216	+0.3260
LSR (L/S)	-0.0161	+0.0110	+0.1352	+0.1670	+0.0076	+0.0004	-0.0415	-0.0226	+0.2410
DMY (kg/plot)	+0.0351	+0.0435	+0.0398	+0.5681	-0.0338	+0.0012	-0.0206	-0.0092	+0.6240
DMC (%)	+0.0122	-0.0111	-0.0035	+0.0653	-0.2942	+0.0013	-0.0115	+0.0075	-0.2340
CP (%)	+0.0128	+0.0401	+0.0166	+0.2091	-0.1262	+0.0031	-0.0056	+0.0380	+0.1880
ADF%	+0.0071	+0.0142	+0.0695	+0.1449	-0.0421	+0.0002	-0.0807	-0.0331	+0.0800
NDF%	+0.0069	+0.0444	+0.0350	+0.0597	+0.0253	-0.0014	-0.0306	-0.0873	+0.0520

Note – PH = plant height, NOT = No. of tillers, LSR = leaf stem ratio, DMY = dry matter yield, DMC = dry matter content, CP = crude protein, ADF = Acid detergent fibre, NDF = Neutral detergent fibre, GFY = Green forage yield

Table 6: Phenotypic Path Coefficients for traits

	PH (cm)	NOT	LSR (L/S)	DMY (kg/plot)	DMC (%)	CP (%)	ADF%	NDF%	GFY (kg/plot)
PH (cm)	+0.0954	+0.0199	-0.0299	+0.2836	-0.0579	+0.0012	-0.0105	-0.0058	+0.2960
NOT	+0.0103	+0.1840	+0.0114	+0.1327	+0.0180	+0.0014	-0.0079	-0.0169	+0.3330
LSR (L/S)	-0.0210	+0.0155	+0.1360	+0.1714	+0.0070	+0.0008	-0.0501	-0.0176	+0.2420
DMY (kg/plot)	+0.0475	+0.0429	+0.0409	+0.5695	-0.0381	+0.0023	-0.0250	-0.0060	+0.6340
DMC (%)	+0.0181	-0.0109	-0.0031	+0.0712	-0.3047	+0.0026	-0.0137	+0.0054	-0.2350
CP (%)	+0.0181	+0.0414	+0.0171	+0.2124	-0.1307	+0.0061	-0.0066	+0.0301	+0.1880
ADF%	+0.0104	+0.0151	+0.0709	+0.1481	-0.0433	+0.0004	-0.0962	-0.0264	+0.0790
NDF%	+0.0080	+0.0453	+0.0348	+0.0501	+0.0241	-0.0027	-0.0370	-0.0687	+0.0540

Note – PH = plant height, NOT = No. of tillers, LSR = leaf stem ratio, DMY = dry matter yield, DMC = dry matter content, CP = crude protein, ADF = Acid detergent fibre, NDF = Neutral detergent fibre, GFY = Green forage yield.

Conclusion

The present investigation revealed significant genotypic variation among 13 oat (*Avena sativa* L.) genotypes for all yield and quality traits studied. PCV was higher than GCV for all traits, with a narrow difference for most characters indicating limited environmental influence. High heritability coupled with appreciable genetic advance as percentage of mean was recorded for ADF% (97.94%, 22.61%) and crude protein (93.13%), indicating effectiveness of direct selection for these quality traits. Dry matter yield (69.16%) and leaf to stem ratio (65.78%) also recorded high heritability with reasonable genetic advance. Among the genotypes, UPO-22-2 was consistently superior for green forage yield (149.17 kg/plot), dry matter yield (33.22 kg/plot), and crude protein content (15.16%). UPO-22-10 and UPO-22-4 were the earliest to flower and mature, desirable for timely forage harvesting, while UPO-212 recorded the lowest ADF (7.15%) and NDF (28.60%), indicating better forage digestibility potential. Dry matter yield showed the strongest positive association with green forage yield at genotypic (r = 0.624**) and phenotypic (r = 0.634**) levels and exerted the highest direct positive effect (0.568) in path analysis, followed by number of tillers, confirming these as the most important selection criteria for forage yield improvement. Crude protein was negatively associated with NDF%, consistent with the known inverse relationship between protein and fibre in forage. Genotypes UPO-22-2, UPO-22-9, and UPO-22-5 were promising for forage productivity while UPO-212 stood out for quality, and may be considered for further evaluation.

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